Get Started with the CDM for Collateral

Module C – How the CDM works

Version 2.0 – Feb 2025

Click below to get started!

Start

Overview

There are six modules in this Get Started guide:

A	Introduction to the CDM	15 minutes	Complete
B	Accessing the CDM	10 minutes	Complete
C	How the CDM works	30 minutes	Start
D	How the CDM is used	45 minutes	Start
E	Collateral Use Cases	45 minutes	Start
F	Advanced Features	30 minutes	Start

How the CDM Works

C	The third module in this series introduces you to some of the key concepts and models in the CDM; you will have an opportunity to experience some hands-on navigation of the CDM.

1	Key concepts: Data Types and Attributes	Start
2	Exploring the data model	Start
3	Legal Agreements	Start
4	Eligible Collateral Specification in the CDM	Start

C1.1 Key concepts: Data Types and Attributes

- The CDM has some very important design principles.
- The first is **normalisation**: the concept of abstraction of common components.
- Second is **composability**: the ability to compose more complex objects from these normalised common components.
- Think of this as **composing** models using common **normalised** toy blocks.

- If you start with the right set of **normalised** building blocks, it is quite easy to **compose** something that represents a model of something in the real world – such as a house:

- The availability of the appropriate basic building blocks is crucial to the success of the model.

Next

Modelling using Composition

- If one has access to the full variety of blocks, and enough of them, one can build a much more complex and sophisticated model:

Next

C1.3 Modelling using Composition

- If one has access to the full variety of blocks, and enough of them, one can build a much more complex and sophisticated model:

- Or even create something completely different which is unrelated to the original vision and purpose of the building blocks:

Next

C1.4 Normalisation and Composition in CDM

- So how do the principles of **normalisation** and **composition** actually work in CDM?
- We call our basic building blocks **data types**.
- In our **normalised** model, we start with a set of basic data types:
  - *int* – for integer numbers
  - *number* – for decimal numbers
  - *boolean* – logical true or false
  - *string* – text
  - *date* – a simple date value
  - *time* – simple time values, e.g. “05:00:00”
- We can then create new data types by **composition** of these basic data types.
- For example, we can create a data type to model a “person” as a composition of a several named basic data types:

```
type Person:
  firstName string
  familyName string
  dateOfBirth date
```

- We can read this as follows:
  - It is a definition of a new Person **data type**
  - Composed of these **attributes:**
    - *firstName* and *familyName*, which are strings
    - *dateOfBirth*, which is a date.

Next

C1.5 Rune DSL

- We can enhance this definition of Person further using some additional modelling features. In so doing, we introduce you to the **Rune DSL** (domain specific language) which is the syntax we use in the CDM model to define components.

- On the right, we have used composition to add the attribute *birthplace* to the Person data type; birthPlace is itself defined as an **Enum** (short for enumerated data type), which has a set of allowed values.

- We have created a new data type Parent which is an inheritance **extension** of Person, i.e. a particular type of person, with its own additional attribute, *role*, which is also defined as a **Enum**.

- We also introduce **cardinality**, which is the number of times an attribute can occur with a lower and upper bound expressed as *(lower..upper).*

```
type Person: <“Defines the person characteristics”>
   firstName string (1..*) <“At least one”>
   familyName string (1..1)
   dateOfBirth date (1..1)
   birthPlace CountryCodeEnum (1..1)

type Parent extends Person: <“Parent is a specific
   type of person, with a specified role”>
   role ParentRoleEnum (1..1)

type Family:
   parents Parent (0..2)  <“Up to two parents”>
   children Person (0..*) <“Zero or more children”>

enum CountryCodeEnum : <“List of Country Codes”>
   France
   Germany
   Japan <“etc...”>

enum ParentRoleEnum: <“List of roles for Parent”>
   Mother
   Father
```

- Comments have also been added, inside angle brackets and quotes.

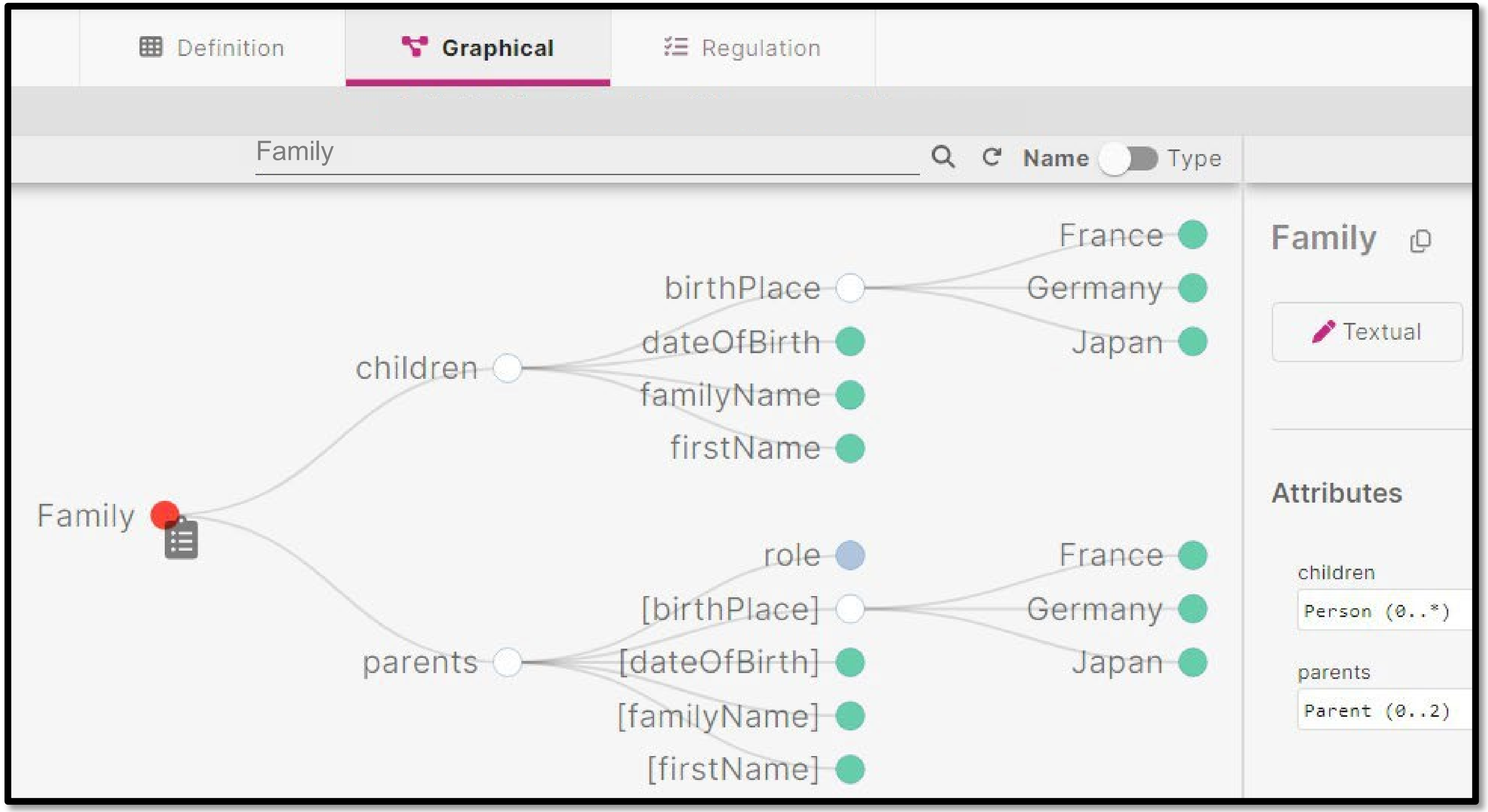
Next

C1.6 Graphical view

- In the previous module, you got access to CDM via Rosetta and saw briefly the Textual and Graphical view.

- Our model of Family was shown as **textual** Rosetta DSL code on the previous page.

- Here is how the Family model is shown in the **graphical** view in Rosetta, which shows the relationship between children and parents and the re-use in composition of the Person data type.



Next

C1.7 Conditions

- Sometimes it is appropriate to be able to define the relationships between data types and attributes.
- A **condition** can be added to a data type definition to define these relationships.
- The condition *ChildrenNeedParents* on the right ensures that a Family object cannot be created with children without at least one parent.
- The condition *ChildrenYoungerThanParents* checks that, if there are children, their birth year/s is after the birth year of the parents.

```
type Family:
parents Parent (0..2) <"Up to two parents">
children Person (0..*) <"Zero or more children">

condition ChildrenNeedParents:
if children exists
then parents count > 0

condition ChildrenYoungerThanParents:
if parents exists
then filter children -> dateOfBirth all > parents -> dateOfBirth max
then True
```

- You can read more details on the syntax of the condition statement by following the link below to the Rune DSL documentation. However, this is not strictly necessary as the examples you will find in this course will not be complex and should be self-explanatory.

[DSL Documentation]

[Done]

How CDM Works

C	The third module in this series introduces you to some of the key concepts and models in CDM.

1	Key concepts: Data Types and Attributes	Complete
2	Exploring the data model	Start
3	Legal Agreements	Start
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C2.1 Exploring the data model

- Recap and objectives of this tutorial:
  - In module B, you got access to the CDM, be it via Rosetta, GitHub or some other means.
  - In the earlier part of this module C, we introduced you to data types, attributes and enums, and the foundational syntax in the Rosetta DSL.
- Next, we will guide you to navigate and explore the CDM further, focusing on the following foundational modules:
  - Product Model
  - Event Model
  - Process Model
  - Reference Data Model
  - Legal Agreements Model.
- This will be a high-level overview before we take a deep dive into the Collateral world in the next section.

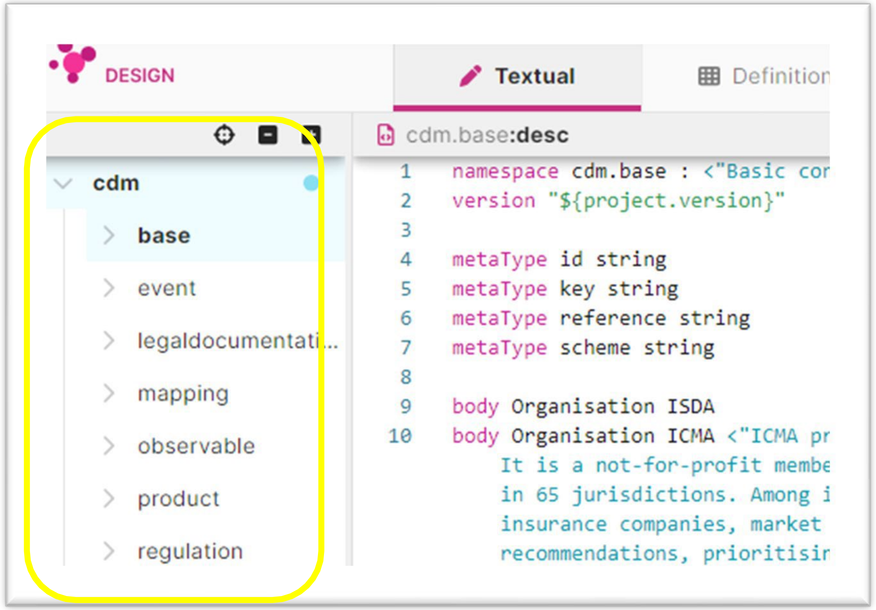
Next

C2.2 **The CDM Models & Namespaces**

- Some of the most important components in the CDM are the model definitions; a set of models, expressed in the Rosetta DSL, which are organized as namespaces.

- Please take a moment to open your favourite tool and make sure you can navigate the CDM to find these models and their corresponding definitions.

- *If using Rosetta:*
  - Open the current production version of the CDM in the Workspace Manager.
  - View the models in the Namespace viewer left panel.



- *If using GitHub or your own tool:*
  - Each of the models exists as a file with a *.rosetta* suffix.
  - They can be found in the following path: `common-domain-model / rosetta-source / src / main / rosetta.`

The main CDM models and Namespaces:

- *Base*
- *Product Model*
- *Event Model*
- *Process Model*
- *Reference Data Model*
- *Legal Agreements Model*

Next

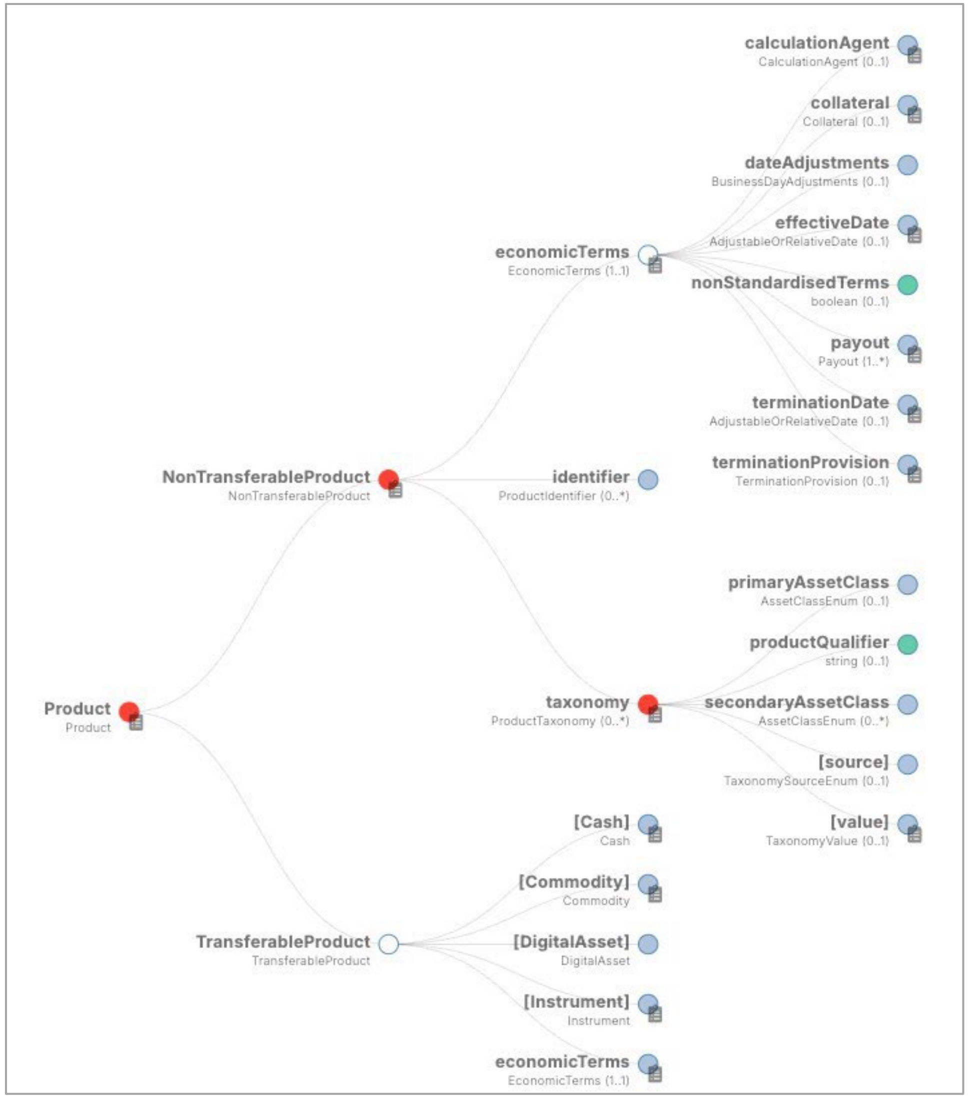
C2.3 The Models in the CDM

- We will explore each of the key Models in the CDM in the following pages. As we do so, we recommend that you follow along by looking at the model yourself and exploring it further.

- Please open your preferred tool now and navigate to the Product definition.

- While there are many ways to do this, we will illustrate the CDM in this guide using the Rosetta tool and both the Textual and Graphical views. For example, here is a view of the Product data type in the graphical view.

- To see this for yourself:
  - Login to Rosetta.
  - Open the CDM production workspace.
  - Navigate to the *Graphical* tab.
  - Enter “Product” in the search bar.

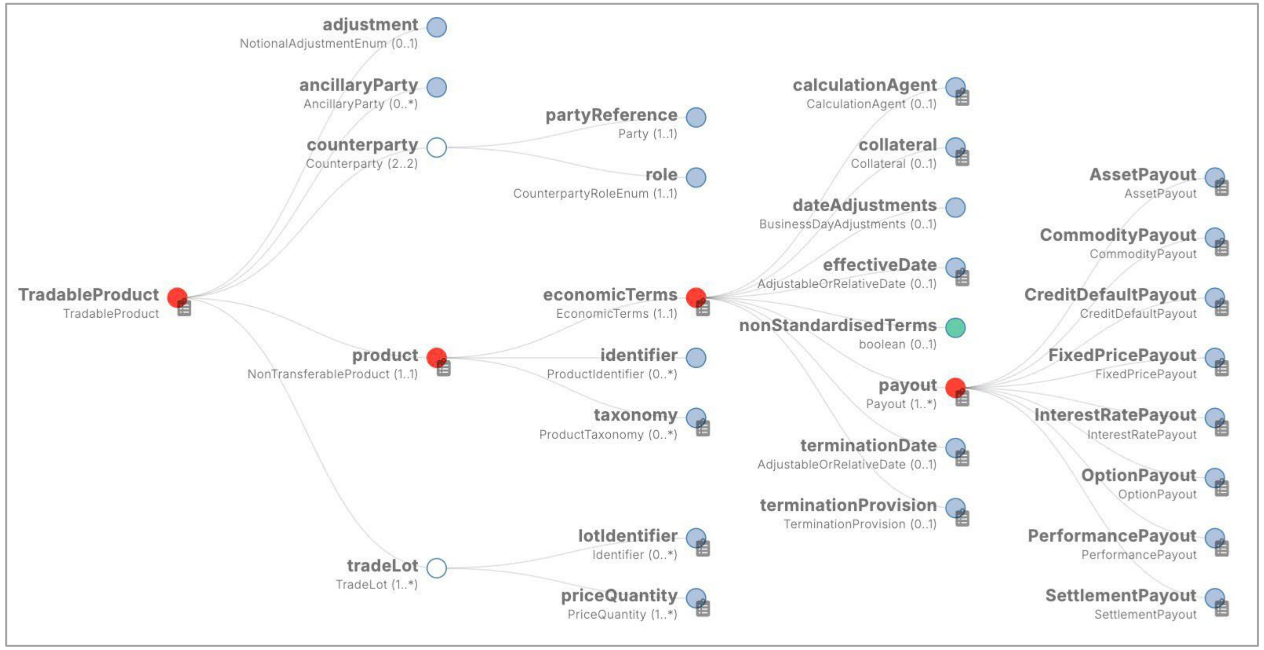


Next

C2.4 The Product Model

- Look at the data type TradableProduct, which is a “definition of a product as ready to be traded, i.e. included in an execution or contract”. TradableProduct includes Product and Party information, but also the tradeLot, which specifies the price and quantities.

- You will see several familiar-looking financial products inside the Product data type and also ContractualProduct which is used for a negotiated trade which has specific terms – i.e., a typical OTC. These terms are digitized inside EconomicTerms whose key attribute is Payout which enables modelling of the legs of a swap, for example a FixedPricePayout and an InterestRatePayout to model a typical interest rate swap.



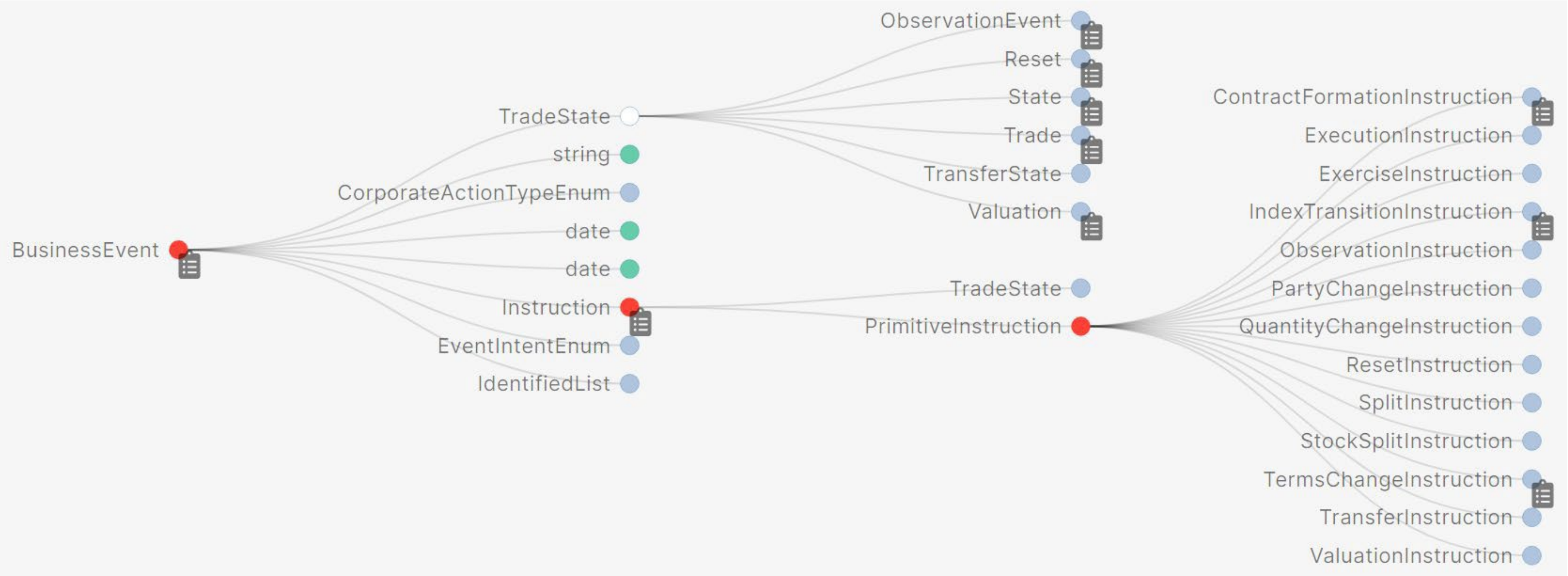
- This is a very high-level overview of the Product Model at this stage but be curious and explore it a bit further!

- **Follow along in your tool:** as you navigate, look at the documentation of each data type

Next

C2.5 The Event Model

- The CDM event model provides data structures to represent the lifecycle events of financial transactions.
- A lifecycle event occurs when a transaction goes through a state transition initiated either by one or both trading parties, by contractual terms, or by external factors. For example:
  - The execution of a trade is the initial event which results in the state of an executed trade.
  - Subsequently, one party might initiate an allocation.
  - Both parties might initiate an amendment to a contractual agreement.
  - Or a default by an underlying entity on a Credit Default Swap would trigger a settlement.



- In the CDM, a BusinessEvent is used to apply an Instruction on a TradeState.
- Search for BusinessEvent and explore the hierarchy beneath it.

Next

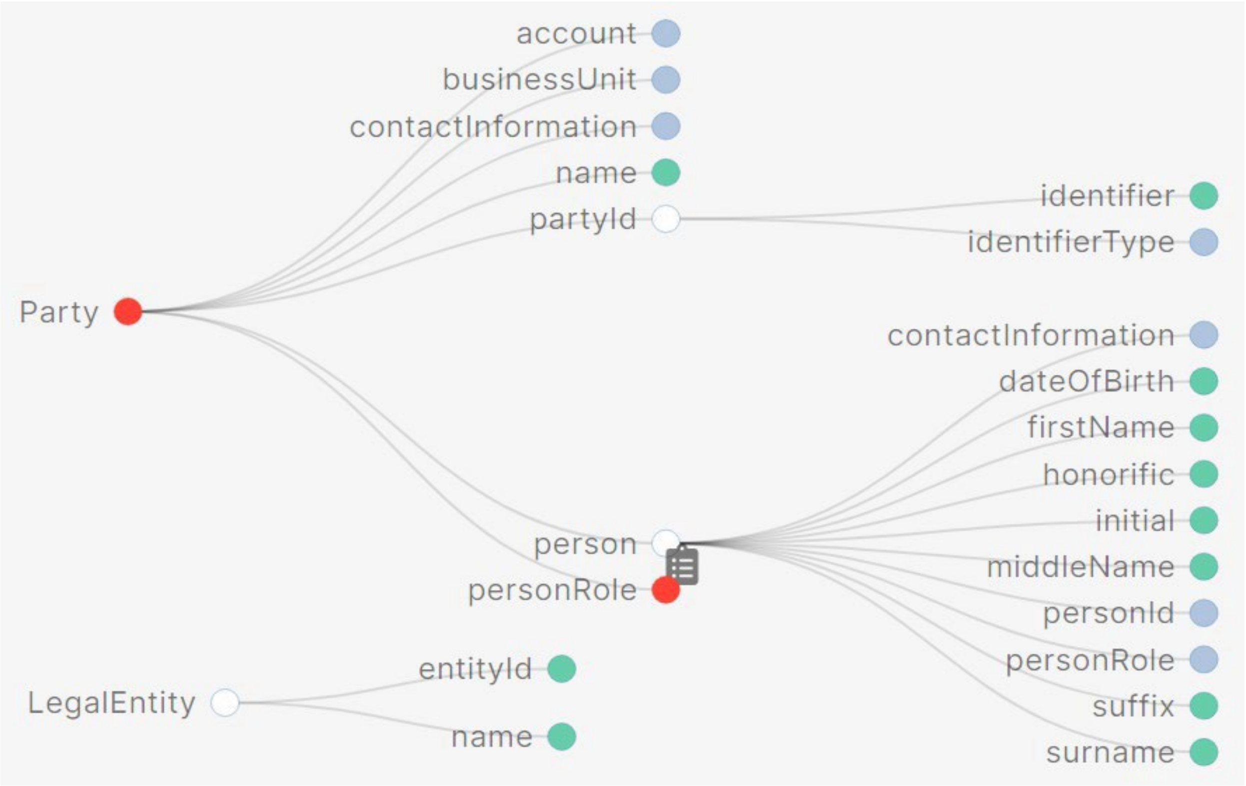
C2.6 The Reference Data Model

- The CDM only integrates the reference data components that are specifically needed by the model; this translates into the representation of Party and LegalEntity.

- Parties are not explicitly qualified as a legal entity or a natural person, although the model provides the ability to associate a person (or set of persons) to a party, which would imply that such a party would be a legal entity.

- The LegalEntity type is used when only a legal entity reference is appropriate i.e. the value would never be that of a natural person.

- Please explore these two data types further in the model.



Done

How CDM Works

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C3.1 The Use of Agreements in Financial Markets

- Financial transactions consist primarily of agreements between parties to make future payments or deliveries to each other.
- To ensure performance, those agreements typically take the form of legally enforceable contracts, which the parties record in writing to minimize potential future disagreements.
- It is common practice in some markets for different aspects of these agreements to be recorded in different documents, most commonly dividing those terms that exist at the trading relationship level (e.g. credit risk monitoring and collateral) from those at the transaction level (the economic and risk terms of individual transactions).

Next

C3.2 The Use of Agreements in Financial Markets

- Relationship agreements and individual transaction level documents are often called "master agreements" and "confirmations" respectively, and multiple confirmations may be linked to a single master agreement.

- Both the relationship and transaction level documents may be further divided into those parts that are standard for the relevant market, which may exist in a pre-defined base form published by a trade association or similar body, and those that are more bespoke and agreed by the specific parties.

- The standard published forms may anticipate that the parties will choose from pre-defined elections in a published form, or create their own bespoke amendments.

Next

C3.3 The ISDA Master Agreement

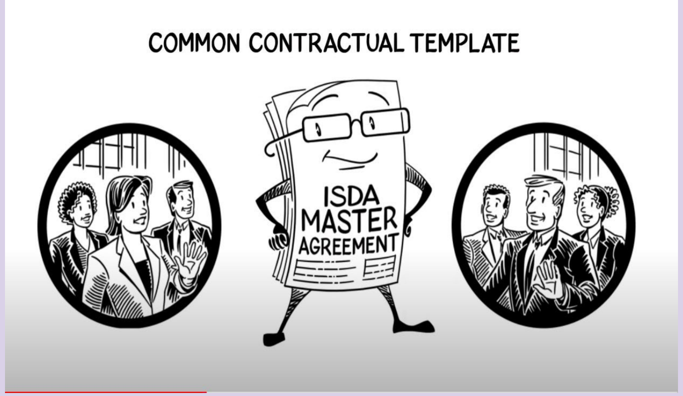
- The ISDA Master Agreement is an internationally recognised document which is used to provide certain legal and credit protection for parties who enter into OTC derivatives.
- Parties that execute agreements for OTC derivatives are expected to have bi-lateral Master Agreements with each other that cover an agreed range of transactions.
- Accordingly in the CDM, each transaction can be associated with a single master agreement, and a single master agreement can be associated with multiple transactions.

Want to refresh your understanding on the ISDA Master Agreement?

Watch this ISDA video:

Understanding the ISDA Master Agreement.

Watch



Next

C3.4 Other types of Agreements

- In addition to the Master Agreement are sets of credit support documentation which parties may enter into as part of Master Agreement to contain the terms on which they will exchange collateral for their OTC derivatives. Collateral provides protection to a party against the risk that its counterparty defaults and fails to pay the amount that it owes on default.

- There are several different types of ISDA credit support documents, reflecting variation and initial margin, regulatory requirements and terms for legal relationships under different legal jurisdictions.
  - Credit Support Annexes (CSAs) exist in various governing jurisdictions such as New York, English, Irish, French, and Japanese law forms. They define the terms for the provision of collateral by the parties in derivatives transactions, and in some cases, they are specialized for initial margin or variation margin.
  - Credit Support Deed (CSD) is very similar to a CSA, except that it is used to create specific types of legal rights over the collateral under English and Irish law, which requires a specific type of legal agreement (a deed).
  - The Collateral Transfer Agreement and Security Agreement (CTA and SA) together define a collateral arrangement where initial margin is posted to a custodian account for use in complying with initial margin requirements. The CTA/SA offers additional flexibility by allowing parties to apply one governing law to the mechanical aspects of the collateral relationship (the CTA) and a different governing law to the grant and enforcement of security over the custodian account (the SA).

Next

C3.5 **Legal Agreements in the CDM**

- In the CDM, “legal agreement” refers to the written terms of a relationship-level agreement, and “contract” refers to the written terms defining an executed financial transaction.

- The benefits of this digital representation are summarized below:

  - Supporting marketplace initiatives to streamline and standardise legal agreements with a comprehensive digital representation of such agreements.

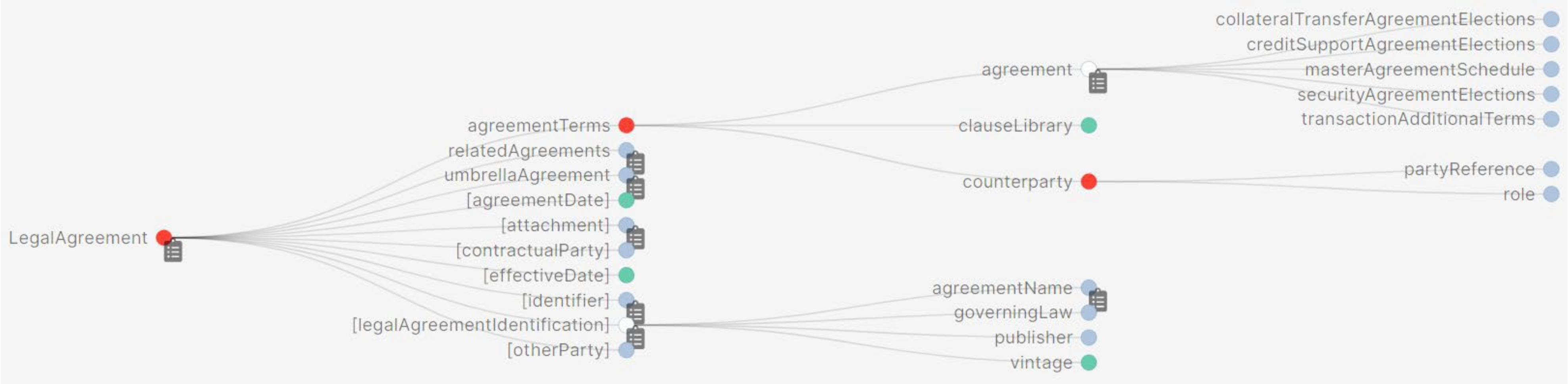
  - Providing a comprehensive representation of the financial workflows by complementing the trade and lifecycle event model and formally tying legal data to the business outcome and performance of legal clauses. (e.g. in collateral management where lifecycle processes require reference to parameters found in the associated legal agreements, such as the Credit Support Annex).

  - Supporting the direct implementation of functional processes by providing a normalised representation of legal agreements as structured data, as opposed to the unstructured data contained of a full legal text that needs to be interpreted first before any implementation (e.g. for a calculation of an amount specified in a legal definition).

Next

C3.6 Legal Agreements in the CDM

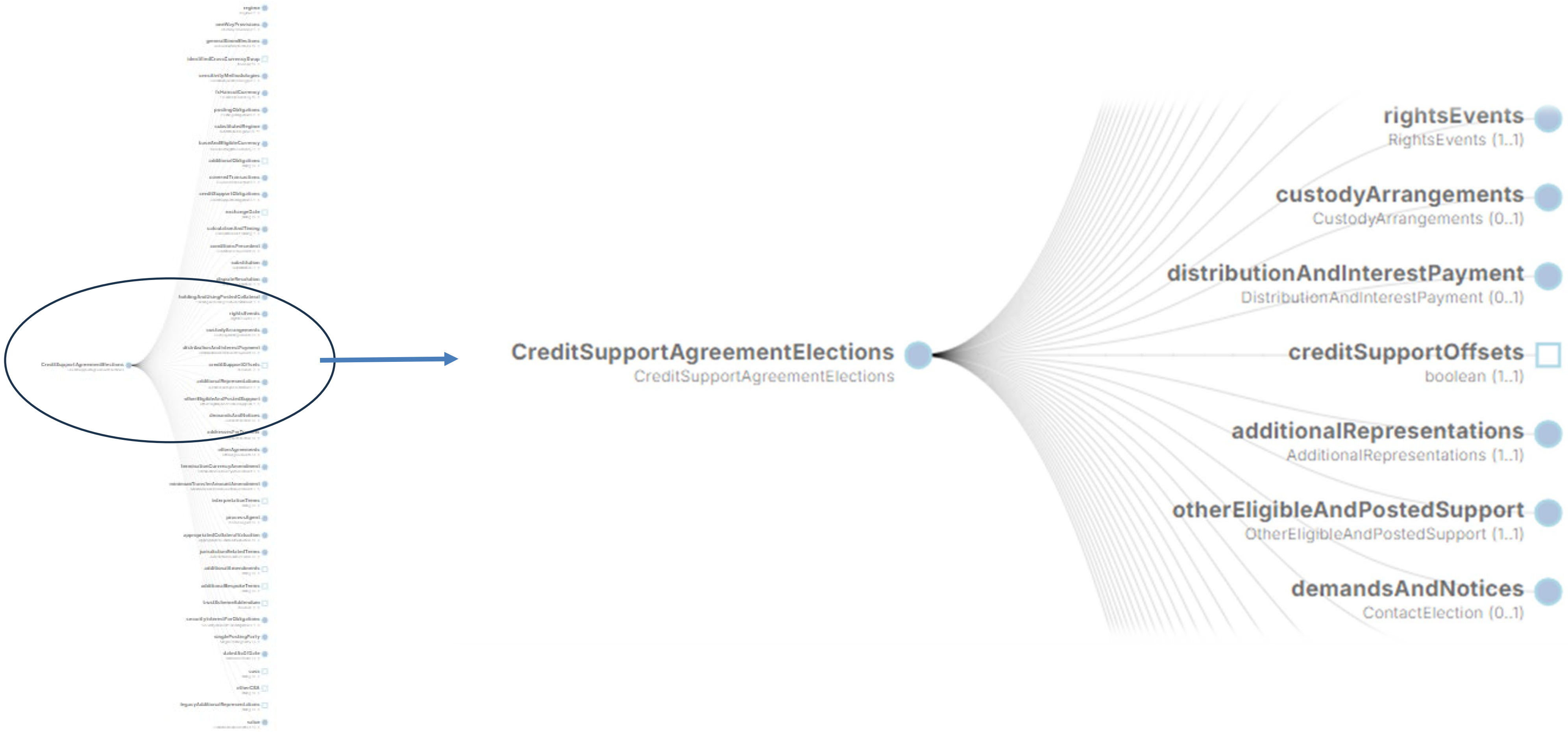
- The LegalAgreement data type represents the highest-level data type for defining a legal agreement in the CDM.
- All legal agreements must contain an agreement date, two contractual parties, and information indicating the published form of market standard agreement being used (including the name and publisher of the legal agreement being specified in the agreementIdentification attribute).
- AgreementTerms is used to specify the content of a legal agreement in the CDM.
  - Agreement is a data type used to specify the individual elections contained within the legal agreement. It contains a set of encapsulated data types, each containing the elections used to define a specific group of agreements.
  - Each counterparty to the agreement is assigned an enumerated value of either Party1 or Party2 through the association of a CounterpartyRoleEnum with the corresponding Party.



Next

C3.7 ISDA Agreements in the CDM

- You can review and click through to the underlying Elections within agreement > agreementTerms for agreement specific elections for each set of agreement type such as CreditSupportAgreementElections (there are more attributes here than can be displayed graphically on a single page).



Done

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1	Key concepts: Data Types and Attributes	Complete
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CDM COMMON DOMAIN MODEL

C4.1 Eligible Collateral Specification in the CDM

- In this section, we are going to drill into the modelling of data for Eligible Collateral.
- Within collateral documentation, it is common to detail what assets you will exchange with your counterparties, i.e., what you deem eligible collateral. Such information is found in bilateral legal documents, custodian triparty agreements, and is also used for other purposes where defining whether an asset is eligible to be used as collateral to mitigate risk on a defined set (portfolio) of financial instruments between parties.
- Data requirements to represent eligible collateral include common information such as asset descriptors e.g. who issues the asset, the asset type, its maturity profile, any related agency credit risk rating, and if any collateral haircut is to be applied to the asset's value.
- Within legal collateral documents, the definition of eligible collateral can take several forms; some may want to list assets and the related eligibility information in table-format using common language, use textual description of types of eligible assets, or use common identifiers and taxonomies. However, it is evident for each method chosen there is no common data standard to express the same information for all the data attributes used.

Need to refresh your understanding on Collateral?

Watch this ISDA video:

How is Collateral Used in the Derivatives Market.

Watch

Next

C4.2 Eligible Collateral Specification in the CDM

- The financial crisis and the resulting regulatory framework that emerged from guidelines outlined under BCBS/IOSCO and Basel III has presented further requirements that define specific criteria for collateral eligibility that must be applied to portfolios.
- Observation of different regulations under various jurisdictions has presented several challenges for defining collateral asset economic identity, correct categorisation, and application of specified haircuts and concentration limits.
- Having no common standards in place to represent the key data has led to lengthy negotiation, misinterpretation, lack of interoperability, and downstream operational inefficiency.
- The CDM provides a standard digital representation of the data required to express collateral eligibility for purposes such as representation in legal agreements that govern transactions and workflows.

In this ISDA video, understand what is being done to automate and standardize collateral processes across market partcipants.

Watch

Next

C4.3 The Benefits of using CDM for Eligible Collateral

The benefits of the digital representation of Eligible Collateral in CDM are summarized below:

- Provides a comprehensive digital representation to support the data requirements to universally identify collateral types.
- Includes ability to identify attributes of collateral that contribute to the risk like the type of asset, interest structures, economics, embedded options and unique characteristics.
- Uses data standards to specify eligibility related information such as haircuts (regulatory credit quality, FX related or additional haircuts), agency or composite credit ratings and asset maturity terms.
- Provides functions to apply treatment rules to predefined collateral criteria such as include/exclude logic.
- Applies treatment rules for concentration limit caps by percentage or value. These can be applied to one or multiple elements of the collateral characteristics and defined criteria.
- Includes attributes to identify regulatory rules by defined eligibility identification categories included in specific rulesets, for example European EMIR requirements or US prudential regulators' rules.
- Provides a means of Identifying Schedules and constructing reusable collateral profiles.
- Standardises digital data representation components to construct the details to identify collateral eligibility not just for regulatory purposes but for all needs of eligibility expression within legal contracts and documentation.
- Promotes a standard format to represent eligible collateral for negotiators to agree details without misinterpretation.
- Provides standards to facilitate Interoperability between platforms for digitised eligible collateral information.
- Connects contractual terms of eligibility in documentation to supporting processes.
- Provides many opportunities in the collateral ecosystem and benefits a data representation of collateral choices that can be imported and exported to other systems such as credit, treasury, trade reporting and custodian platforms, providing a full workflow solution from negotiation, execution through to optimisation and settlement.

Next

C4.4 Modelling Approach and Scope

- Although initially built to meet uncleared margin rules in OTC Derivatives, the model is now cross-product and extensible to many different use cases.

- In addition, the design can express collateral eligibility for other industry workflows such as Securities Lending, Repo and Exchange Traded Derivatives (ETD). The model foundations, broad range of attributes and functions have been constructed with this in mind and can be extended further to operate to wider processes.

- The common data requirements have been established through industry working groups reviewing a wide range of examples to identify collateral for the purpose of constructing eligible collateral specifications, including representation of additional attributes for regulatory risk and credit factors. For the purpose of understanding the principle, these can be divided into the following categories:

Issuer Identification	Asset Identification	Collateral Haircuts	Maturity Range	Concentration Limits
Treatment Functions	Agency Ratings	Exchanges	Indexes	Industry Sectors

- The data attributes within the model provide the flexibility to identify the collateral criteria, including attributes of the issuer and asset class, then define its maturity if relevant, then apply treatment rules for any chosen haircut percentages, concentration limits and inclusion or exclusion conditions.

- A combination of these terms allows a wide range of collateral and associated eligibility data to be represented.

Next

Approach to identifying Collateral Assets

- Historically, there have been no common standards for describing the assets used as collateral; the foundational structure in the CDM provides a means to identify a majority of collateral issuers and covers a wide range of asset types that are commonly seen in eligible collateral data.

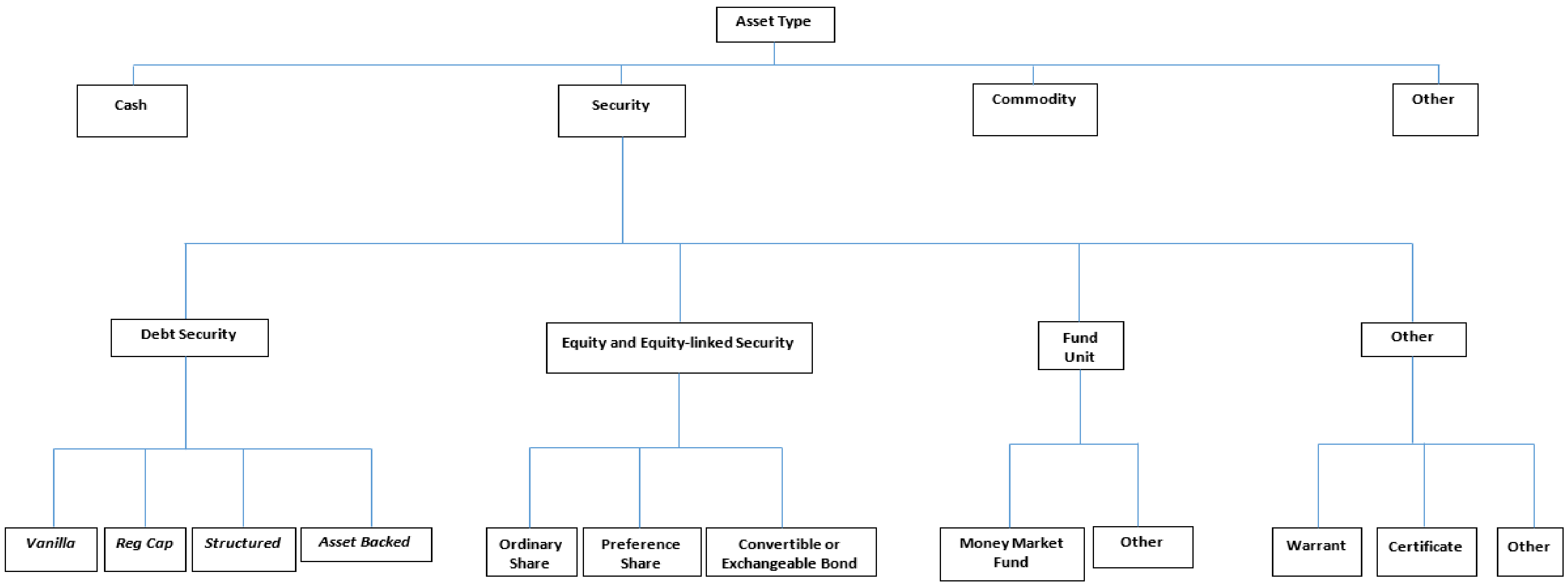
- The intention is for the CDM to provide a standard means of identifying as much of this collateral universe as possible initially and then extend the model further as required.

- The CDM method can represent common master and respective collateral documentation for OTC Derivatives, like the ISDA CSA, and non-OTC master agreements (notably Repo and Securities Lending) and potentially for OTC Cleared and Exchange Traded Derivatives.

Next

C4.6 Approach to identifying Collateral Assets

- Assets are modelled using a hierarchical Asset Type tree, shown here.

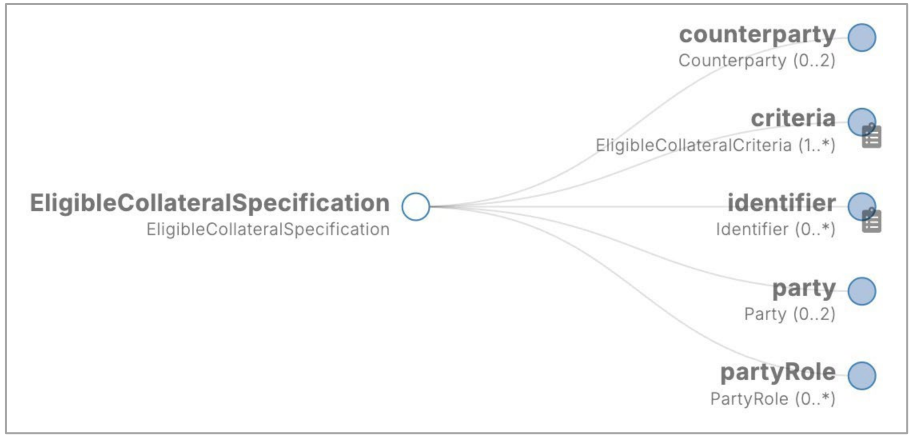


- Eligible collateral can also be represented using industry identifiers, where available:
  - Collateral issuers can be identified using common legal entity identifiers such as an LEI.
  - Assets can be identified using a product ID such as ISIN or CUSIP or a standard taxonomy source.

Next

C4.7 The Eligible Collateral Specification

- It is time to look at the CDM again; please open your preferred tool and access the current production version of the CDM.
- The highest level and foundational data structure for the representation of eligibility is the EligibleCollateralSpecification which is a root class; search for and open this data type now.
- An EligibleCollateralSpecification typically represents the schedule of eligible collateral agreed between two parties and is represented digitally as one or more EligibleCollateralCriteria to define the details.
- Navigate the model and read the definitions and comments to confirm your understanding.



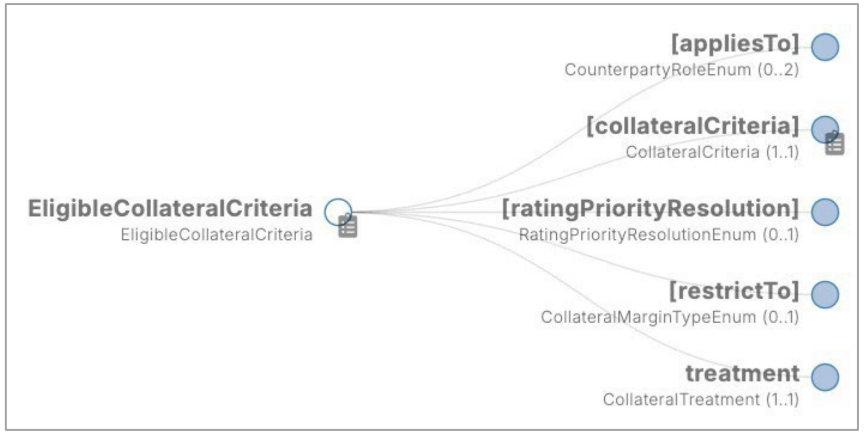

```
type EligibleCollateralSpecification: <"Represents a set of criteria used
    to eligible collateral.">
    [rootType]
    [metadata key]

    identifier Identifier (0..*) <"Specifies the identifier(s) to
    uniquely identify eligible collateral or a set of eligible
    collateral, such as a schedule or equivalent for an identity Issuer.">
    party Party (0..2) <"The parties associated with the specification.">
    counterparty Counterparty (0..2) <"Specification of the roles of the
    counterparties to the specification.">
    criteria EligibleCollateralCriteria (1..*) <"Represents a set of
    criteria used to specify eligible collateral.">
    partyRole PartyRole (0..*) <"Specifies the role(s) that each of the party(s) is playing in the context of
the specification, e.g. Payor or Receiver.">
```

Next

C4.8 The Eligible Collateral Criteria

- The collateralCriteria attribute is used to specify and describe a given criteria for collateral, such as the type of asset and its maturity characteristics, and issuer details and agency rating
- The appliesTo attribute is used to specify which of the two counterparties the criteria applies to (either one or both counterparties).
- The treatment attribute is used to specify the valuation percentage, any concentration limits and/or specific inclusion or exclusion conditions, which additionally apply to filter whether a collateral item is eligible or not.
- A combination of these terms allows a wide variety of eligible collateral types to be represented.




```
type CollateralCriteriaBase: <"Represents a set of criteria used to specify and
describe collateral.">

    collateralCriteria CollateralCriteria (1..1) <"The specific criteria that
    applies. It can be created using AND, OR and NOT logic, and both asset and
    issuer characteristics.">
    appliesTo CounterpartyRoleEnum (0..2) <"Specifies which of the two
    counterparties the criteria applies to (either one or both counterparties).
    This attribute is optional, in case the applicable party is already specified
    elsewhere within a party election.">
    restrictTo CollateralMarginTypeEnum (0..1) <"Restrict the criteria to only
    apply to a specific type of margin, ie IM or VM.">
    ratingPriorityResolution RatingPriorityResolutionEnum (0..1) <"Denotes which
    Criteria has priority if more than one agency rating applies.">

type EligibleCollateralCriteria extends CollateralCriteriaBase: <"Represents a set
of criteria used to specify eligible collateral.">

    treatment CollateralTreatment (1..1) <"Identifies the treatment of
    specified collateral, e.g., haircuts, holding limits or exclusions.">
```

Next

C4.9 AND, OR, and NOT Logic for CollateralCriteria

- collateralCriteria also allows CDM users to combine different attributes using AND, OR, and NOT logic to model the eligibility schedule fully

```
type AllCriteria: <"Used to combine two or more Collateral Criteria using AND logic.">
allCriteria CollateralCriteria (2..*)

type AnyCriteria: <"Used to combine two or more Collateral Criteria using OR logic.">
anyCriteria CollateralCriteria (2..*)

type NegativeCriteria: <"Used to apply a NOT logic condition to a single Collateral
Criteria.">
negativeCriteria CollateralCriteria (1..1)
```

- For example, a schedule may specify that either of the following would qualify as eligible collateral:
  - Equity assets from US issuers
  - Bond instruments from UK issuers.
- This can be described logically as:
- (AssetType = 'Equity' AND IssuerCountryOfOrigin = 'USA') OR (AssetType = 'FixedIncome' AND IssuerCountryOfOrigin = 'UK’).

- AllCriteria, AnyCriteria, and NegativeCriteria can be used to create simple collateral schedules limited to specific criteria or can be nested and used in tandem to create more complex schedules to account for multiple conditions.

Next

C4.10 Additional Details

- Now that you have become familiar with the basic construct for Collateral, please explore the model further, focusing on gaining an understanding of the following data types:
  - AssetType
  - AssetMaturity
  - CollateralIssuerType
  - CollateralTreatment including
    - valuationTreatment
    - concentrationLimit
  - CollateralValuationTreatment including
    - haircutPercentage
    - marginPercentage
  - Inclusion Rules using isIncluded
  - AgencyRatings.

- Please also read the CDM documentation on Eligibilility Collateral to enhance your understanding: cdm.finos.org/docs/eligible-collateral-representation/
- We suggest that you read all of the following sections:
  - Introduction.
  - Eligible Collateral in the CDM.
  - Modelling Approach.
  - Using the CDM to identify Eligible Collateral.
  - Additional Granular Construction.

Next

C4.11 Summary

- The CDM data structure to express collateral eligibility has been explored in more detail and it has been demonstrated where the EligibleCollateralCriteria can be broken down into data related to CollateralCriteria and rules can be applied using data for CollateralTreatment.
- In the next module we will look at specific use cases in the collateral industry and how the CDM can be used to address them.

How the CDM Works

C	You have completed the third module.

1	Key concepts: Data Types and Attributes	Complete
2	Exploring the data model	Complete
3	Legal Agreements	Complete
4	Eligible Collateral Specification in the CDM	Complete
	Take your knowledge check	Go

- You have completed the third module of the Getting Started guide to using the CDM in Collateral.
- To help you confirm your understanding, we have a few questions to check your knowledge and understanding.
- If you would like to go back over any of the materials, please click "Review" to return to the module menu and then select the sections you would like to read again.
- For some of these questions, you will need access to the current production version of the CDM, so open it now and be ready to explore the model to answer the questions.
- When ready, please click "Start" to check your knowledge.

C5.1 How the CDM Works: Knowledge Check

Q: Which of the following statements is NOT true?

A1 Composability is the ability to create more complex objects from normalised common components

A2 Normalisation is the concept of abstraction of common components

A3 The CDM cannot be extended to create new models

A4 The keyword *extends* denotes inheritance from another data type

A5 The Rune DSL supports normalisation and composability

Q: Which of the following statements is not true?

A1 Composability is the ability to create more complex objects from normalised common components

A2 Normalisation is the concept of abstraction of common components

A3 The CDM cannot be extended to create new models

A4 The keyword extends denotes inheritance from another data type

A5 The Rune DSL supports normalisation and composability

The CDM is designed to be infinitely extensible to support new models for future financial products and markets, even those that have not yet been created.

The CDM design is able to do this by being built on normalised components which can be used to create new models using composition and extension.

Next

C5.3 How the CDM Works: Knowledge Check

Q: A CollateralCriteria data type can be used to define the issuers and/or assets that meet a given criteria for collateral. Which of the following is NOT supported?

A1 The Type of Asset

A2 The Issuer Agency Rating

A3 The Currency Code

A4 The Maturity Band of the Asset

A5 The Collateral Interest Calculation Parameters

CDM
COMMON DOMAIN MODEL

C5.4 How the CDM Works: Knowledge Check

Q: A CollateralCriteria data type can be used to define the issuers and/or assets that meet a given criteria for collateral. Which of the following is NOT supported?

A1 The Type of Asset

A2 The Issuer Agency Rating

A3 The Currency Code

A4 The Maturity Band of the Asset

A5 The Collateral Interest Calculation Parameters

The parameters for calculating interest on a collateral item is not defined in CollateralCriteria

However, it can be defined in the CollateralInterestCalculationParamters.

Next

Q: Which of the following statements is NOT true about MaturityTypeEnum?

A1 It contains the value “RemainingMaturity”

A2 It is used within AssetCriteria

A3 It represents an enumeration list to identify the Maturity

A4 It contains the value “OriginalMaturity”

A5 None of the above

How the CDM Works: Knowledge Check

Q: Which of the following is statements is NOT true about MaturityTypeEnum?

A1 It contains the value “RemainingMaturity”

A2 It is used within AssetCriteria

A3 It represents an enumeration list to identify the Maturity

A4 It contains the value “OriginalMaturity”

A5 None of the above

Statements 1, 2, 3 and 4 are all true, therefore the only one that is not true is 5.

```
type AssetCriteria:
 collateralAssetType AssetType (0..*)
 assetCountryOfOrigin ISOCountryCodeEnum (0..*)
 denominatedCurrency CurrencyCodeEnum (0..*)
 agencyRating AgencyRatingCriteria (0..*)
 maturityType MaturityTypeEnum (0..1)

enum MaturityTypeEnum: <"Represents an
  enumeration list to identify the Maturity.">
  RemainingMaturity
  OriginalMaturity
  FromIssuance
```

Next

CDM COMMON DOMAIN MODEL

C5.7 How the CDM Works: Knowledge Check

Q: Which of the following is NOT a benefit of the digital representation of Eligible Collateral in CDM?

A1	Provides functions to apply treatment rules to predefined collateral criteria such as include/exclude logic
A2	Provides standards to facilitate interoperability between platforms for digitised eligible collateral information
A3	Provides a comprehensive digital representation to support the data requirements to universally identify collateral types
A4	Provides an online solution to automate the creation, negotiation and execution of derivatives agreements
A5	Provides a means of Identifying Schedules and constructing reusable collateral profiles

C5.8 How the CDM Works: Knowledge Check

Q: Which of the following is NOT a benefit of the digital representation of Eligible Collateral in CDM?

A1 Provides functions to apply treatment rules to predefined collateral criteria such as include/exclude logic

A2 Provides standards to facilitate interoperability between platforms for digitised eligible collateral information

A3 Provides a comprehensive digital representation to support the data requirements to universally identify collateral types

A4 Provides an online solution to automate the creation, negotiation and execution of derivatives agreements

A5 Provides a means of identifying Schedules and constructing reusable collateral profiles

An online solution that helps to automate derivatives agreements, including Master Agreements, is actually solutions such as **ISDA Create**. This leverages CDM but is not explicitly related to the Eligible Collateral model.

The other answers are all correct and are some of the many benefits of using CDM to model Eligible Collateral.

Next

C5.9 How the CDM Works: Knowledge Check

Q: Which of the following is NOT a basic type in the Rosetta DSL?

A1 date

A2 integer

A3 float

A4 string

A5 boolean

Q: Which of the following is NOT a basic type in the Rosetta DSL?

A1 date

A2 integer

A3 float

A4 string

A5 boolean

"float" is not a Rosetta DSL basic data type.

The "number" basic type is used to represent a decimal number, such as 3.14. "integer" is used for whole numbers.

"number" can be parameterized to support different precisions, eg a number with 3 decimal places:

```
number (fractionalDigits: 3)
```

Next

C5.11 How the CDM Works: Knowledge Check

Q: Which of these is NOT an attribute of the EconomicTerms data type?

A1 payout

A2 collateral

A3 terminationDate

A4 agreementDate

A5 calculationAgent

C5.12

How the CDM Works: Knowledge Check

Q: Which of these is NOT an attribute of the EconomicTerms data type.?

A1 payout

A2 collateral

A3 terminationDate

A4 agreementDate

A5 calculationAgent

"agreementDate" is not an attribute of the EconomicTerms data type.

There is an attribute on EconomicTerms, named effectiveDate, that records when the terms apply.

agreementDate itself appears as an attribute on the data type LegalAgreement.

Next

C5.13 How the CDM Works: Knowledge Check

Q: Looking in the CDM at the *conditions* defined on the data type EligibleCollateralCriteria, which of the following statements is most accurate?

A1 They define the use of Concentration Limits

A2 They define the use of Security Types

A3 They are restricted to be used only for certain assets

A4 They exclude the use of Concentration Limit Types only

A5 They identify the treatment of specified collateral

How the CDM Works: Knowledge Check

Q: Looking in the CDM at the *conditions* defined on the data type EligibleCollateralCriteria, which of the following statements is most accurate?

A1 They define the use of Concentration Limits

A2 They define the use of Security Types

A3 They are restricted to be used only for certain assets

A4 They exclude the use of Concentration Limit Types only

A5 They identify the treatment of specified collateral

All three conditions define the use of concentration limits, specifically:

- IssueOutstandingAmount can only be used with Security and Debt assets.
- MarketCapitalisation can only be used with Security and Equity assets.
- AverageTradingVolume can only be used with Security and Equity assets.

Next

How the CDM Works: Knowledge Check

Q: A CollateralValuationTreatment will allow for representation of different types of haircuts. Which of the following is NOT a supported haircut type?

A1 fxHaircutPercentage

A2 levelPercentage

A3 haircutPercentage

A4 marginPercentage

A5 AdditionalHaircutPercentage

How the CDM Works: Knowledge Check

Q: A CollateralValuationTreatment will allow for representation of different types of haircuts. Which of the following is NOT a supported haircut type?

A1	fxHaircutPercentage
A2	levelPercentage
A3	haircutPercentage
A4	marginPercentage
A5	AdditionalHaircutPercentage

A levelPercentage is not a valid haircut type.

levelPercentage is used to trigger a payment event.

Done

COMMON DOMAIN MODEL

How the CDM Works

C	1	Key concepts: Data Types and Attributes	Review
	2	Exploring the data model	Review
	3	Legal Agreements	Review
	4	Eligible Collateral Specification in the CDM	Review
	5	Take your knowledge check	Review

Well Done! You have completed the third module
You can review any of the complete sections again or move on to the next module.

Done